```
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'Project 6
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' {$STAMP BS2}
' {$PBASIC 2.5}

x           VAR Word
y           VAR Word
row         VAR Nib
column      VAR Nib
keypad      VAR Word
temp        VAR Nib
position1   VAR Word
position2   VAR Word
Dball       VAR Word
Dball2      VAR Word
start_game  VAR Word
game_mode   VAR Word
counter     VAR Word
position3   VAR Word
Dball3      VAR Word

counter = 0
position1 = 129
Dball = 5

SEROUT 14, 84, [22, 12, 17]
SEROUT 14, 84, [208]
SEROUT 14, 84, [248, %11111, %11111, %11111, %11111, %11111, %11111, %11111, %11111]
SEROUT 14, 84, [249, %11111, %11111, %10000, %10000, %10000, %10000, %10000, %11111]
SEROUT 14, 84, [250, %11111, %11111, %10001, %10001, %10001, %10001, %10001, %10001]
SEROUT 14, 84, [251, %00110, %00110, %00000, %00000, %00000, %00000, %00000, %00000]
SEROUT 14, 84, [252, %00000, %00000, %00110, %00110, %00000, %00000, %00000, %00000]
SEROUT 14, 84, [253, %00000, %00000, %00000, %00000, %00110, %00110, %00000, %00000]
SEROUT 14, 84, [254, %00000, %00000, %00000, %00000, %00000, %00000, %00110, %00110]
SEROUT 14, 84, [255, %11111, %11111, %10000, %10000, %10110, %10110, %10000, %11111]

GOTO start

start:
      counter = 0
      game_mode = 0
      SEROUT 14, 84, [12, 17]
      PAUSE 10
      SEROUT 14, 84, [133, "Maze Game"]
      SEROUT 14, 84, [153, "(1) Easy", 173, "(2) Hard", 193, "(*) Start"]
      DO
      GOSUB ReadKeypad
      IF keypad THEN
      GOSUB display
      ENDIF
      LOOP
      RETURN


wrong:
      SEROUT 14, 84, [12, 17]
      SEROUT 14, 84, [128, "Wrong number/letter", 148, "Please try again"]
      PAUSE 1500
      GOTO start
      RETURN

layout:
      SEROUT 14, 84, [12, 17]
      SEROUT 14, 84, [128, 1, " ", 0, " ", " ", " ", 0, " ", " ", " ", 0, " ", " ", " ", 0, " ", " ",
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" ", 0, 2]
      SEROUT 14, 84, [148, 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0,
 " "]
      SEROUT 14, 84, [168, 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0, " ", 0,
 " "]
      SEROUT 14, 84, [188, 0, " ", " ", " ", " ", 0, " ", " ", " ", " ", 0, " ", " ", " ", " ", 0, " ", " ", " ", " ", 0,
 " ", " ", " "]
      IF counter = 0 THEN
      SEROUT 14, 84, [128, 7]
      ENDIF
      RETURN

game:
      DO
      PULSIN 13, 1, x
      PULSIN 12, 1, y
      x = x / 125
      y = y / 125
      PAUSE 50
      IF (counter = 0) THEN
      DO
      PULSIN 13, 1, x
      PULSIN 12, 1, y
      x = x / 125
      y = y / 125
      IF (x > 21) THEN
      position1 = 129
      Dball = 5
      SEROUT 14, 84, [128, 1, 5]
      SEROUT 14, 84, [218, 225]
      counter = 1
      PAUSE 500
      position2 = position1
      Dball2 = Dball
      position2 = 128
      Dball3 = 3
      GOTO game
      ENDIF
      LOOP
      ENDIF

      IF (position1 = 129 OR position1 = 131 OR position1 = 132 OR position1 = 133 OR position1 =
134 OR position1 = 136 OR position1 = 137 OR position1 = 138 AND Dball = 3 AND y < 19) THEN
      GOTO lose
      ENDIF
      IF (position1 = 140 OR position1 = 141 OR position1 = 142 OR position1 = 144 OR position1 =
145 OR position1 = 146 AND Dball = 3 AND y < 19) THEN
      GOTO lose
      ENDIF

      IF (position1 = 189 OR position1 = 190 OR position1 = 191 OR position1 = 193 OR position1 =
194 OR position1 = 195 OR position1 = 197 OR position1 = 198 AND Dball = 6 AND y > 21) THEN
      PAUSE 50
      GOTO lose
      ENDIF
      IF (position1 = 199 OR position1 = 201 OR position1 = 202 OR position1 = 203 OR position1 =
205 OR position1 = 206 OR position1 = 207 AND  Dball = 6 AND y > 21) THEN
      PAUSE 50
      GOTO lose
      ENDIF

      IF (x > 21) THEN
      position3 = position2
      position2 = position1
      SEROUT 14, 84, [position1, " "]
      SEROUT 14, 84, [218, 225]
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      position1 = position1 + 1
      PAUSE 250
      ELSEIF (x < 19) THEN
      position3 = position2
      position2 = position1
      SEROUT 14, 84, [position1, " "]
      SEROUT 14, 84, [218, 222]
      position1 = position1 - 1
      PAUSE 250
      ENDIF
      IF (y > 21) THEN
      Dball3 = Dball2
      Dball2 = Dball
      IF Dball = 3 THEN
      Dball = 4
      PAUSE 250
      ELSEIF Dball = 4 THEN
      Dball = 5
      PAUSE 250
      ELSEIF Dball = 5 THEN
      Dball = 6
      PAUSE 250
      ELSEIF Dball = 6 THEN
      Dball = 3
      SEROUT 14, 84, [position1, " "]
      position3 = position2
      position2 = position1
      position1 = position1 + 20
      PAUSE 250
      ENDIF
      SEROUT 14, 84, [218, 225]
      ENDIF
      IF (y < 19) THEN
      Dball3 = Dball2
      Dball2 = Dball
      IF Dball = 3 THEN
      Dball = 6
      SEROUT 14, 84, [position1, " "]
      position3 = position2
      position2 = position1
      position1 = position1 - 20
      PAUSE 250
      ELSEIF Dball = 4 THEN
      Dball = 3
      PAUSE 250
      ELSEIF Dball = 5 THEN
      Dball = 4
      PAUSE 250
      ELSEIF Dball = 6 THEN
      Dball = 5
      PAUSE 250
      ENDIF
      SEROUT 14, 84, [218, 228]
      ENDIF

      IF (position1 = 130 OR position1 = 134 OR position1 = 138 OR position1 = 142 OR position1 =
146 OR position1 = 148 OR position1 = 150 OR position1 = 152 OR position1 = 154 OR position1 = 156)
THEN
      GOTO lose
      ELSEIF (position1 = 158 OR position1 = 160 OR position1 = 162 OR position1 = 164 OR position1 =
 166 OR position1 = 168 OR position1 = 170 OR position1 = 172 OR position1 = 174) THEN
      GOTO lose
      ELSEIF (position1 = 176 OR position1 = 178 OR position1 = 180 OR position1 = 182 OR position1 =
 184 OR position1 = 186 OR position1 = 188) THEN
      GOTO lose
      ELSEIF (position1 = 192 OR position1 = 196 OR position1 = 200 OR position1 = 204) THEN
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      GOTO lose
      ENDIF

      IF (position1 = 147) THEN
      GOTO win
      ENDIF
      SEROUT 14, 84, [8]
      SEROUT 14, 84, [position1, Dball]

      IF (game_mode = 2) THEN
      IF (position1 = position3) THEN
      GOTO lose
      ELSEIF (Dball = Dball3) THEN
      IF (y > 21 OR y < 19) THEN
      PAUSE 10
      ELSE
      GOTO lose
      ENDIF
      ENDIF
      ENDIF
      LOOP
      RETURN

lose:
      SEROUT 14, 84, [12, 17]
      SEROUT 14, 84, [150, "Loser Try Again!!"]
      PAUSE 200
      SEROUT 14, 84, [218, 227]
      PAUSE 100
      SEROUT 14, 84, [218, 226]
      PAUSE 100
      SEROUT 14, 84, [218, 225]
      PAUSE 100
      GOTO start
      RETURN

win:
      SEROUT 14, 84, [12, 17]
      SEROUT 14, 84, [150, "You Won Smarty!!"]
      PAUSE 200
      SEROUT 14, 84, [218, 220]
      PAUSE 100
      SEROUT 14, 84, [218, 223]
      PAUSE 100
      SEROUT 14, 84, [218, 228]
      PAUSE 100
      GOTO start
      RETURN

ReadKeypad:
      keypad = 0
      OUTL = %00000000
      DIRL = %00000000
      FOR row = 0 TO 3
      DIRB = %1111
      OUTB = %0000
      OUTA = 1 << row
      DIRA = 1 << row
      temp = 0
      FOR column = 0 TO 3
      INPUT (column + 4)
      temp = temp | (INB & (1 << column))
      NEXT
      keypad = keypad << 4 | (Temp REV 4)
      NEXT
      RETURN
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Display:
      IF KeyPad.BIT15 THEN game_mode = 1
      IF Keypad.BIT14 THEN game_mode = 2
      IF KeyPad.BIT13 THEN GOTO wrong
      IF Keypad.BIT12 THEN GOTO wrong
      IF KeyPad.BIT11 THEN GOTO wrong
      IF Keypad.BIT10 THEN GOTO wrong
      IF KeyPad.BIT9 THEN GOTO wrong
      IF Keypad.BIT8 THEN GOTO wrong
      IF KeyPad.BIT7 THEN GOTO wrong
      IF Keypad.BIT6 THEN GOTO wrong
      IF KeyPad.BIT5 THEN GOTO wrong
      IF Keypad.BIT4 THEN GOTO wrong
      IF KeyPad.BIT3 THEN
      IF (game_mode = 1 OR game_mode = 2) THEN
      SEROUT 14, 84, [209, 218, 220]
      SEROUT 14, 84, [209, 222]
      SEROUT 14, 84, [209, 223]
      SEROUT 14, 84, [209, 225]
      SEROUT 14, 84, [209, 227]
      GOSUB layout
      GOTO game
      ELSE
      GOTO wrong
      ENDIF
      IF Keypad.BIT2 THEN GOTO wrong
      IF KeyPad.BIT1 THEN GOTO wrong
      IF Keypad.BIT0 THEN GOTO wrong
      ENDIF
      RETURN
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